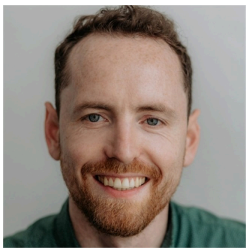
Dr. John D. Kirwan

Data Scientist · Statistical Modelling · Machine Learning · Life Sciences

La Laguna, Spain • jkirwan.org • Contact form



KEY SKILLS

Python · R · pandas · Git · PyTorch · Stan · Statistical Modelling · Bayesian Statistics · Experimental Design · Data Visualisation · Machine Learning · Docker

PROFILES

JohnKirwan
johndkirwan
0000-0001-5537-3574
johnkirwan

LANGUAGES

English (Native)
Spanish (B2)
Swedish (B2)

PROFILE

Data scientist with a PhD and a background in computational biology, combining rigorous statistical modelling — Bayesian inference, experimental design and uncertainty quantification — with practical machine learning in Python and R. Nearly four years in industry R&D building data pipelines, training and validating deep-learning models, and translating results for experimental scientists, including a contribution to a NeurIPS 2025 paper on uncertainty quantification.

SELECTED HIGHLIGHTS

- Uncertainty-aware modelling: Bayesian multilevel models (R, Stan) and deep-learning uncertainty quantification
- Experimental design and power analysis, from behavioural experiments to preclinical studies
- Data products: open-source R packages with interactive Shiny apps (radiatR on CRAN, luxR)

RECENT EXPERIENCE

Data Scientist / Bioinformatician *Freelance, La Laguna, Spain* (Jul 2026-present)

Bioinformatics Scientist *Arquimea Research Centre, La Laguna, Spain* (Jul 2022-Mar 2026)
Worked at the interface of the AI and biotech teams on structure-based drug discovery, generative molecular design, and applied machine learning — the link between the computational tools and the biology.

- Contributing author on a NeurIPS 2025 paper on uncertainty quantification for deep regression: helped develop the method in PyTorch and ran the empirical validation
- Built a modular synthetic-data-generation pipeline (Python, Biopython) orchestrating GNINA, DiffDock and AlphaFold 2, including dataset sourcing and inter-tool format conversion
- Predicted structures and binding affinities for protein and protein–peptide complexes, including metal-ion and ATP cofactors, and designed de novo peptides, using Boltz, BoltzGen, AlphaFold, Foldseek and BioEMU
- Developed the binding-affinity objective for the multi-objective small-molecule optimisation engine of Uncharted, a venture that spun out as Uncharted Chem
- Applied AI tools and data analysis to cancer-research and gut-microbiota (metagenomic) programmes; prototyped a novel operon-design approach for extremophile bacteria and ran statistical power analyses for preclinical studies
- Ran GPU workloads on an NVIDIA DGX cluster and AWS; Docker, Git and Azure DevOps within an Agile/Scrum team

Postdoctoral Researcher *Stazione Zoologica Anton Dohrn, Naples, Italy* (Oct 2019-Jul 2022)
Quantitative sensory biology within an international Human Frontier Science Program (HFSP) collaboration, with emphasis on statistical modelling and signal processing of behavioural data.

- Analysed experimental data using Bayesian multilevel regression models in R and Stan
- Designed and executed behavioural experiments with automated data acquisition pipelines
- Contributed to interdisciplinary publications spanning biology, optics, and computation

EDUCATION

PhD Biology (Vision Science) *Lund University, Lund, Sweden* (May 2013-Jun 2018) Thesis: ‘Spatial Vision in Diverse Invertebrates’.

MSc by Research Biology (Molecular Evolution) *University College Dublin, Dublin, Ireland* (Sep 2008-Sep 2010) Thesis: ‘The Molecular Evolution of Hearing in Mammals’

Dr. John D. Kirwan Data Scientist

ADDITIONAL EXPERIENCE

Project Manager (contract) *Lund University, Lund, Sweden* (Jun 2019-Aug 2019) Short-term research management contract supervising experimental data collection and a masters student, bridging doctoral and postdoctoral positions.

Doctoral Researcher *Lund University, Lund, Sweden* (May 2013-Jun 2018) Computational and statistical analysis of visual systems in invertebrates, combining experimental biology with quantitative modelling.

- Built statistical models using Bayesian multilevel regression in R and Stan; additional analysis in Python and Matlab
- Developed optical and geometric models to estimate spatial resolution theoretically
- Designed novel experimental protocols for quantifying animal visual performance

ADDITIONAL EDUCATION

BSc (Hons) Zoology *University College Dublin, Dublin, Ireland* (Sep 2004-Jun 2008)

- First Class Honours; UCD Zoology Medal (2008)

OPEN-SOURCE SOFTWARE

- **radiatR** — CRAN R package and Shiny app for circular-arena tracking data; imports from DeepLabCut, SLEAP, EthoVision and other trackers, circular statistics and figures
- **luxR** — R package and Shiny app for modelling underwater light environments, photoreceptor excitation and visual contrast

AWARDS & CERTIFICATIONS

- Deep Learning with PyTorch — IBM — 2026
- Pharmaceutical Bioinformatics & Applied Pharmaceutical Bioinformatics — Uppsala University — 2020 Pass with Distinction
- Customising your models with TensorFlow 2 — Imperial College London — 2025
- Valohai MLOps Fundamentals Certification — Valohai — 2025
- Applied Software Engineering Fundamentals Specialization — IBM — 2023 Covers Git, GitHub, Linux, and Python, and the fundamentals of software engineering.
- Writing in the Sciences — Stanford University (online) — 2019 Statement of Accomplishment with Distinction
- Mountain Leader Award — Mountain Leader Training Scotland — 2011 Professional certification to lead hillwalking groups.

SELECTED PUBLICATIONS

Uncertainty Quantification for Deep Regression using Contextualised Normalizing Flows Adriel Sosa Marco, John D. Kirwan, Alexia Toumpa, Simos Gerasimou *The Thirty-ninth Annual Conference on Neural Information Processing Systems, 2026* www.proceedings.com/content/085/085713-1692open.pdf

Phenotypic response to food availability in sea urchin larvae and impact of light during development and growth Maria Cocurullo, John D. Kirwan, Maria Ina Arnone *Frontiers in Ecology and Evolution, 2025* www.frontiersin.org/journals/ecology-and-evolution/articles/10.3389/fevo.2025.1548208/full

A Model of Decentralized Vision in the Sea Urchin Diadema africanum Tianshu Li, John D. Kirwan, Maria Ina Arnone, Dan-Eric Nilsson, Giancarlo La Camera *iScience, 2023* www.cell.com/iscience/fulltext/S2589-0042(23)00372-3